How accurate is altitude from pressure geolocators? A global assessment of three pressure-to-altitude conversions on the ground and in flight

Raphaël Nussbaumer[1*] and Mathieu Gravey[2]

[1*]Swiss Ornithological Institute, Seerose 1, Sempach, 6204, Switzerland.

[2]Institute for Interdisciplinary Mountain Research, Austrian Academy of Sciences, Innsbruck, Austria.

*Corresponding author(s). E-mail(s): raphael.nussbaumer@vogelwarte.ch;

Contributing authors: mathieu@mgravey.com;

Abstract

Background: Pressure geolocators record atmospheric pressure throughout the year on small birds, and pressure sensors also provide vertical records in tags equipped with satellite positioning. The altitudes derived from these records reveal migration strategies and inform conservation assessments, including exposure to wind turbines. However, pressure-to-altitude conversions make assumptions about atmospheric conditions whose consequences for altitude error remain poorly quantified.

Methods: We evaluated three conversions: a fixed standard atmosphere, a conversion using local ERA5 surface conditions, and an ERA5 conversion additionally accounting for virtual temperature and a climatological temperature lapse rate from GPT3. Treating reference pressures as exact conversion inputs, we compared estimates with 1,794 GHCNh weather stations worldwide and IGRA2 radiosondes from 629 stations up to 5 km above ground. We summarize absolute errors by their medians and 95th percentiles, distinguish station bias from within-station

standard deviation, and use flight heights from 203 tracked birds to provide ecological context.

Results: At ground-reference stations, the standard atmosphere produced a median absolute error of 38.1 m, with 5% of absolute errors exceeding 140 m. Local ERA5 conditions reduced these to 3.7 m and 15 m, respectively. At 2,400 m above ground, the 90th percentile of tracked flight heights, the 95th percentile of absolute flight-height error fell from 201 m with the standard atmosphere to 69 m with the corrected ERA5 conversion. There, virtual temperature and GPT3 reduced the median station bias of the ERA5 conversion from -30.9 m to -6.4 m.

Conclusions: Local atmospheric conditions substantially reduce the large altitude errors introduced by a fixed standard atmosphere; virtual temperature and a climatological lapse rate further reduce systematic flight-height errors. The corrected conversion needs only measured pressure, its time and an approximate position, and is available in GeoPressureR. Remaining atmospheric and sensor errors must still be considered when interpreting flight behaviour or classifying flights relative to turbine heights.

Keywords: biologging, barometric altimetry, flight altitude, ERA5, radiosonde, multi-sensor geolocator, GeoPressureR

1 Background

Altitude is a central dimension of how birds use their environment [1]. In flight, it helps explain how migrants adjust their flight heights in relation to winds, temperature and visibility [2–4], how aerial insectivores' foraging heights vary with atmospheric conditions [5], and how birds exploit rising air to reduce flight costs [6]. Synchronized changes in flight height can identify birds travelling together, as demonstrated in bee-eaters [7]. Vertical records also document twilight ascents in swifts [8], exploratory climbs during migration [9], and pronounced day–night changes in flight altitude during barrier crossings [10–12]. On the ground, elevation describes habitat use and movements such as elevational commuting, uphill roosting and seasonal migration [13–15]. Flight-height data also inform conservation assessments, including exposure to wind turbines [16, 17] and the effects of artificial light at night on migratory flight height [18].

Lightweight multi-sensor geolocators equipped with pressure sensors make it possible to study the vertical movements of small birds throughout their annual cycle [19–21]. Their low power requirements allow pressure to be recorded over long deployments, and pressure sensors also complement Global Positioning System (GPS) tracking [6, 16, 22]. In both cases, interpreting the records requires knowing how accurately pressure can be converted into altitude, and whether the resulting uncertainty is small enough to resolve the movements or height differences of interest [23].

Pressure-to-altitude conversion requires a reference pressure at a known altitude and an estimate of the temperature profile between that level and the bird. The simplest approach assumes a fixed standard atmosphere [24], as implemented in pamlr [25], and continues to be used in recent geolocator studies [9, 21, 26–28]. To account for weather-driven pressure changes, studies have used nearby weather-station observations [5, 15]. Reanalysis provides reference conditions across space and time, allowing this correction to be applied along migratory tracks; Meier et al. [8], for example, used reanalysis sea-level pressure while retaining a standard temperature profile. Local surface temperature can also be incorporated, as implemented in GeoPressureR using the fifth-generation European Centre for Medium-Range Weather Forecasts reanalysis (ERA5) surface pressure, temperature and elevation [29, 30], and applied in subsequent geolocator studies [12, 17]. Schaub et al. [31] further refined the temperature profile by estimating a lapse rate between the model surface and an upper-air pressure level, requiring additional temperature and geopotential fields from the ERA5 pressure-level dataset.

Previous evaluations have established the performance of pressure-derived heights in controlled device trials [32], at grounded birds [31] and against GPS during flight [33], and have illustrated the benefit of reanalysis correction [30]. In GPS-tracked birds, behavioural classification can identify periods at a known height, such as floating at sea level, that provide calibration references [16, 33]. Comparable ground-level references cannot be reliably established along geolocator tracks, where uncertain positions limit

the assignment of local ground elevation. For long geolocator deployments, however, a geographically broad comparison is needed to establish how atmospheric conversion errors vary with climate, terrain and flight height. Evaluating alternative conversions against common reference observations would quantify the gains from local weather information and temperature-profile refinements, while distinguishing persistent bias from temporal variability.

Here, we compare three pressure-to-altitude conversions: a fixed standard atmosphere, an ERA5 surface reference with the standard lapse rate, and the same reference with virtual temperature and a climatological lapse rate from the Global Pressure and Temperature 3 (GPT3) model [34]. Using worldwide weather stations and radiosonde profiles up to 5 km above the ground, we quantify atmospheric conversion errors on the ground and in flight, distinguish systematic bias from temporal variability, and relate flight-height errors to the heights at which tracked birds fly. Our aim is to establish how much local atmospheric information and a more realistic temperature profile improve a conversion suitable for long pressure records.

2 Methods

2.1 Three conversions from pressure to altitude

Pressure decreases with height because each layer of air supports the weight of the air above it. Combining this hydrostatic approximation with the ideal gas law gives the hypsometric relation [35]:

$$z = z_0 + \frac{R\bar{T}_v}{g\,M} \ln \frac{p_0}{p}, \tag{1}$$

with z the altitude (m) at pressure p, z_0 and p_0 the altitude and pressure of a reference level, $g = 9.80665$ m s^{-2}, $R = 8.31432$ J mol^{-1} K^{-1}, $M = 0.0289644$ kg mol^{-1}, and $\bar{T}_v$ the mean virtual temperature.

The virtual temperature $T_v = T(1 + 0.608\, q)$, with T the air temperature in kelvin and q the specific humidity, accounts for moist air being lighter than dry air. Temperature can vary with height; the conversion uses the mean virtual temperature between the reference level and the bird, weighted by increments of log pressure:

$$\bar{T}_v = \frac{1}{\ln(p_0/p)} \int_p^{p_0} T_v(p')\, d\ln p'.$$

To simplify the computation, temperature is assumed to change at a constant rate with height, giving a linear profile $T(z) = T_0 + L(z - z_0)$, where T_0 is the reference temperature in kelvin. This rate, called the lapse rate $L = dT/dz$ (K m^{-1}), is negative when temperature decreases with height. Without a humidity correction, $T_v = T$, and integration gives the barometric equation:

$$z = z_0 + \frac{T_0}{L}\left[\left(\frac{p}{p_0}\right)^{-RL/(gM)} - 1\right]. \tag{2}$$

The three conversions differ in their reference level and assumed temperature profile (Figure 1):

1. **Standard atmosphere.** The reference is sea level ($z_0 = 0$, $p_0 = 1013.25$ hPa) with $T_0 = 288.15$ K (15 °C) and $L = -6.5$ K km^{-1} [24]. It needs no external data but ignores changes in pressure, temperature and humidity.
2. **ERA5, standard lapse rate.** The ERA5 reanalysis provides surface pressure p_0, 2 m air temperature T_0 and surface geopotential Φ at the location and hour of each measurement, with reference altitude $z_0 = \Phi/g$. This accounts for local pressure and temperature variations, while retaining a constant lapse rate of -6.5 K km^{-1} and no humidity correction.
3. **ERA5, virtual temperature and GPT3 lapse rate.** This conversion adds two refinements to the ERA5 reference. First, humidity at the reference level is

accounted for by replacing T_0 in Equation 2 with the reference-level value of T_v. The specific humidity q needed for this correction is calculated from ERA5 surface pressure p_0 and 2 m dewpoint T_d, via vapour pressure e [36]:

$$q = \frac{0.622\, e}{p_0 - 0.378\, e}, \qquad e = 611.2 \exp\left(\frac{17.67\, T_d}{T_d + 243.5}\right),$$

where T_d is in °C and p_0 and e are in Pa. Second, the standard lapse rate is replaced with a location- and season-dependent value from GPT3, an empirical troposphere model that provides a lapse rate for any location and time of year [34]. Values are interpolated bilinearly on the 1° grid. This formulation does not resolve humidity aloft.

All three conversions are implemented in R. Both ERA5 conversions are available in the GeoPressureR function `pressure_to_altitude()` (`altitude_formula = "standard"` or `"virtual"`). ERA5 single-level variables [37] are read from the European Centre for Medium-Range Weather Forecasts (ECMWF) analysis-ready, cloud-optimized (ARCO) archive at the nearest 0.25° grid cell and hour.

2.2 Reference datasets

We test pressure-to-altitude conversion in two settings: at weather stations with reported elevations, and during balloon ascents, where measured pressure can be compared with the reported height profile. Figure 2 shows their geographic coverage and station elevations.

2.2.1 Ground-level reference: GHCNh

For the ground comparison, we use the National Oceanic and Atmospheric Administration's (NOAA) Global Historical Climatology Network hourly archive (GHCNh)

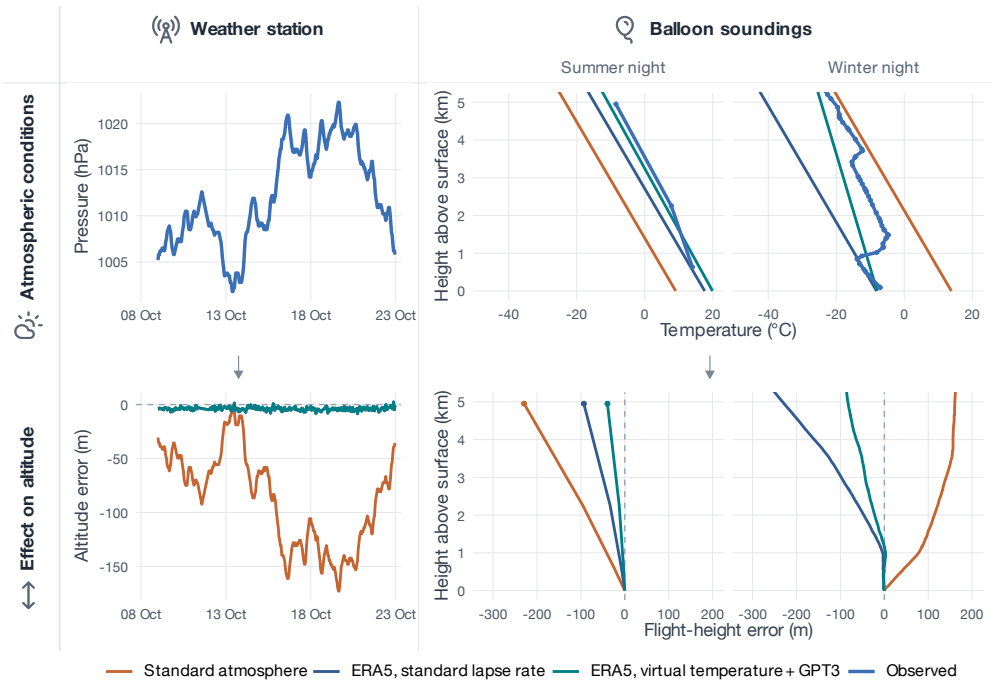


Figure 1: Effects of atmospheric conditions on pressure-derived altitude. Top: observed pressure at Cairo Regional Airport, USA (left), and observed and assumed temperature profiles for a temperate summer night and a continental winter night (right); points mark measured temperature levels, and the ERA5 virtual + GPT3 profile shows virtual temperature. Bottom: the resulting altitude errors at the weather station and flight-height errors above the sounding launch sites. Dashed lines mark zero error.

[38], which provides station pressure and reported elevation. Within 200 km equal-area sampling cells, we select one station using its latest qualifying 12-month window in 2022–2026. We screen reported elevations against independent terrain data and documented reference conflicts, apply archive quality checks, remove isolated pressure spikes and reject years containing large pressure jumps. We also require each station's barometer height, implied by its own sea-level pressure or altimeter reports, to agree with its reported elevation, and its semi-diurnal pressure tide to be consistent with Coordinated Universal Time (UTC) timestamps. This retains 1,794 stations. Each retained report is paired with ERA5 conditions from the nearest hour; selection thresholds are described in Additional file 1: S1.1.

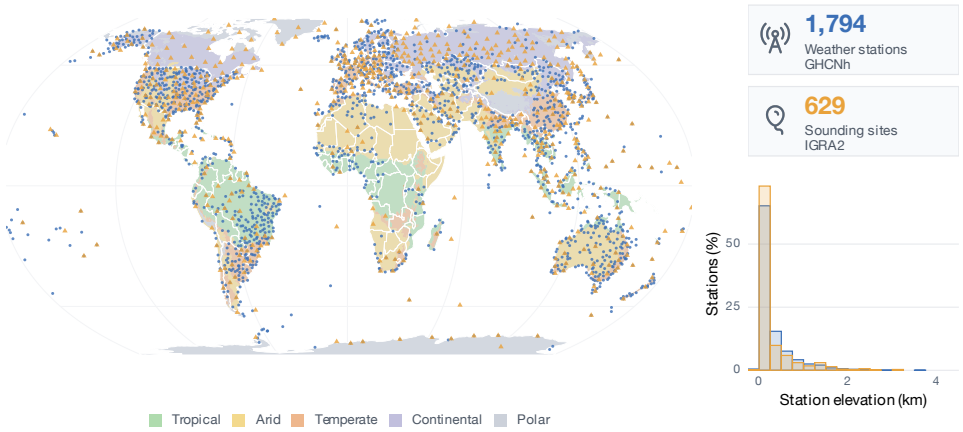


Figure 2: Reference sites for the ground and flight-height comparisons: weather stations (blue circles) and sounding sites (orange triangles), over the main Köppen–Geiger climate zones [42]. The histogram shows station elevations in 250 m bins as a percentage of each network.

2.2.2 Flight-height reference: IGRA2

To assess how conversion error changes above the surface, we use balloon soundings from NOAA's Integrated Global Radiosonde Archive, version 2 (IGRA2) [39–41]. Radiosondes measure pressure as they rise and report the corresponding geopotential height. We compare pressure-derived height changes with reported geopotential-height changes from launch to 5 km above the surface, using 629 stations selected at most one per 200 km equal-area cell. We retain valid surface–airborne pressure–height pairs from soundings whose reported height gains agree with their own temperature profiles, reject station years with more than 10% inconsistent soundings, and require at least 15 launch dates in each month of the latest qualifying 12-month window within 2022–2026 (Additional file 1: S1.2). Selection uses the soundings alone. Each profile level is paired with ERA5 conditions from the nearest hour.

2.3 Error metrics

We calculate height from the measured pressure using each of the three conversions, then subtract the reference height to obtain the error. We summarize individual absolute errors by their median and 95th percentile. Station-level bias (mean signed error) and standard deviation (SD) distinguish persistent offsets from variability. We report median absolute station bias on the ground and median signed station bias in flight, retaining the direction of flight-height errors.

On the ground, we examine how errors vary with terrain roughness, represented by ERA5's standard deviation of sub-grid orography. We also test whether a finer surface reference improves the conversion by replacing ERA5 single-level conditions with ERA5-Land [43], a 0.1° land reanalysis, in the standard-lapse-rate conversion, comparing both products over the same station hours. Supplementary analyses assess the station–ERA5 elevation gap, season, local time and climate (Additional file 1: S2).

In flight, we assess errors in height above the sounding's launch surface, summarizing individual absolute errors and station bias and SD at each height. To relate these errors to bird flight, we use the flight-time height distribution of 203 tagged birds of 25 species from the Master GeoLocator Data Package [44] (Figure 4 c; Additional file 1: S4). We report errors at the heights nearest its median and 90th percentile, and an average over the whole distribution in which the errors in each height band are weighted by the band's share of flight time, with flight time above 5 km assigned to the highest band. To examine how flight-height errors depend on atmospheric conditions, we summarize mean signed error by main Köppen–Geiger climate zone [42], half-year (summer: 21 March–22 September in the northern hemisphere, shifted by six months in the southern hemisphere) and time of day (day: 07:00–19:00 local solar time at launch).

We assess position sensitivity by displacing the atmospheric query location while holding pressure measurements and reference heights fixed (Additional file 1: S5).

2.4 Use of artificial intelligence

OpenAI Codex assisted with R code development, manuscript drafting and revision, and submission formatting. The authors are responsible for checking the code, numerical results, references and manuscript text, and for the scientific interpretation and final content.

3 Results

3.1 Ground altitude errors

Across 1,794 retained GHCNh stations, the median / 95th percentile of absolute hourly errors fell from 38.1 / 140 m with the standard atmosphere to 3.7 / 15 m with ERA5 and the standard lapse rate (Figure 3). The median station |bias| and SD fell from 25.4 and 51.6 m, respectively, to 2.9 and 3.3 m with ERA5. Adding virtual temperature and the GPT3 lapse rate made little difference to ground errors.

ERA5 station offsets and variability were larger in rougher terrain: median |bias| rose from 2.6 m in flat terrain (sub-grid ERA5 orography SD below 20 m) to 5.9 m in mountains (above 300 m); station SD rose from 3.0 to 7.2 m.

Replacing ERA5 single-level conditions with ERA5-Land did not improve ground altitude (Additional file 1: S3). At the 1,500 stations with inputs from both products, the median hourly absolute error was 9.9 m with ERA5-Land and 3.7 m with ERA5, and ERA5-Land errors were larger in 78% of hours. Median |bias| was 8.8 m with ERA5-Land and 2.9 m with ERA5, and ERA5-Land offsets were larger at 79% of stations. Within-station SD was also slightly larger with ERA5-Land (4.4 versus 3.3 m).

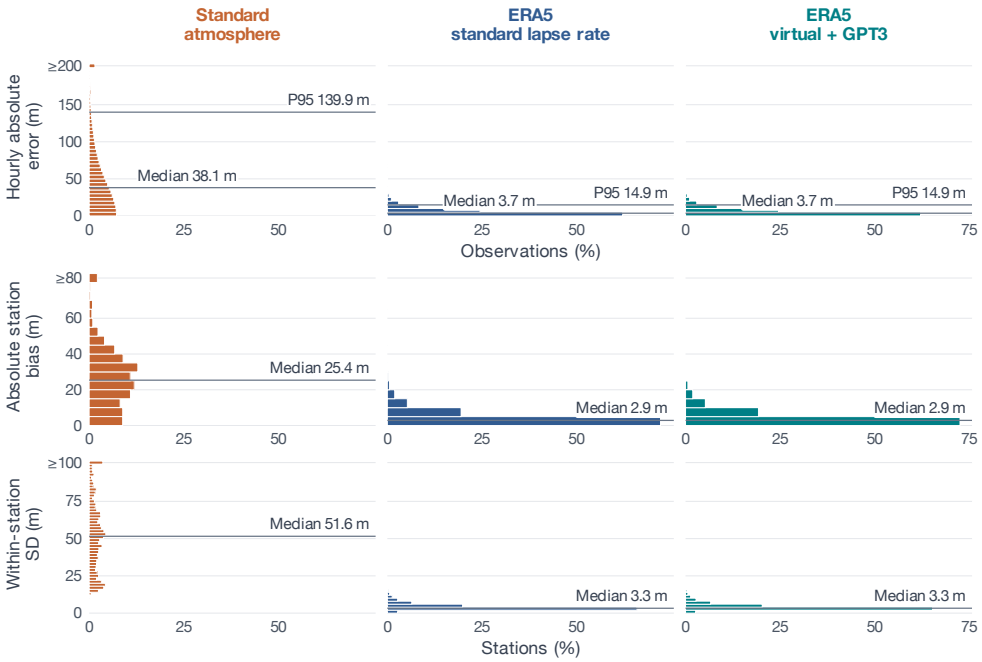


Figure 3: Ground errors of the three conversions (columns): hourly absolute error, absolute station bias and within-station SD (rows). Each station has equal total weight in the hourly distribution. Horizontal lines mark medians and, for hourly errors, the 95th percentile. Axes are shared within each row; the top bins include all values at or above 200 m, 80 m and 100 m, respectively.

3.2 Flight-height errors

In the tracked birds from the Master GeoLocator Data Package, half of recorded flight time was at estimated heights below 582 m and 90% was below 2,363 m (Figure 4 c; Additional file 1: S4). At 600 m above the sounding surface, near the median flight height, the median absolute error was 25.5 m with the standard atmosphere, 6.1 m with ERA5 and the standard lapse rate, and 4.7 m with virtual temperature and GPT3. At 2,400 m, near the 90th percentile, these errors were 102.5, 29.8 and 21.2 m, with 95th percentiles of 201, 97 and 69 m. At both heights, the median absolute error was about 4% of height with the standard atmosphere and 1% with virtual temperature and GPT3, and the 95th percentile about 9% and 3%. Averaged over the

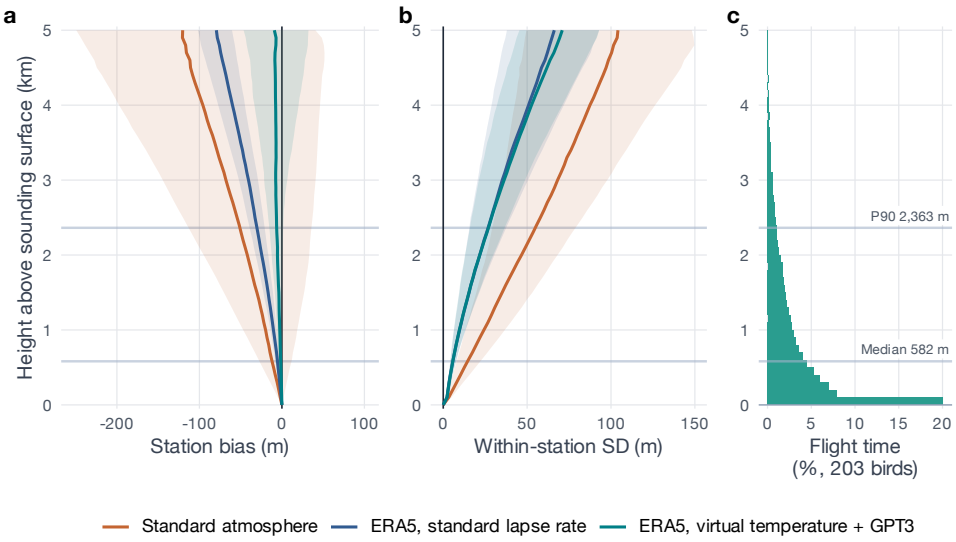


Figure 4: Flight-height errors against height above the sounding surface: (a) station bias (negative: height underestimated) and (b) within-station SD. Lines: medians across stations; shading: 25th–75th percentiles across stations. (c) Flight-time distribution of height above median terrain for 203 tracked birds (Additional file 1: S4); its median and 90th percentile are marked across the panels.

whole flight-time distribution, which is dominated by low flight, the median absolute error was 20.2, 6.5 and 4.9 m.

Station bias and within-station SD both grew with height (Figure 4 a, b). At 2,400 m, adding virtual temperature and GPT3 shifted the median station bias from -30.9 to -6.4 m, while the median station SD remained similar (27.2 and 27.4 m). Displacing the location at which atmospheric conditions are queried by 50 km changed the median absolute error of the corrected conversion near 2 km by less than 2 m (Additional file 1: S5).

Flight-height errors depended on climate, season and time of day (Figure 5). At 2,400 m, the mean error of the standard atmosphere ranged from -130 m in tropical summers to +165 m in polar winters, as its fixed temperature profile is too cold for warm columns and too warm for cold ones. ERA5 surface conditions with the standard lapse

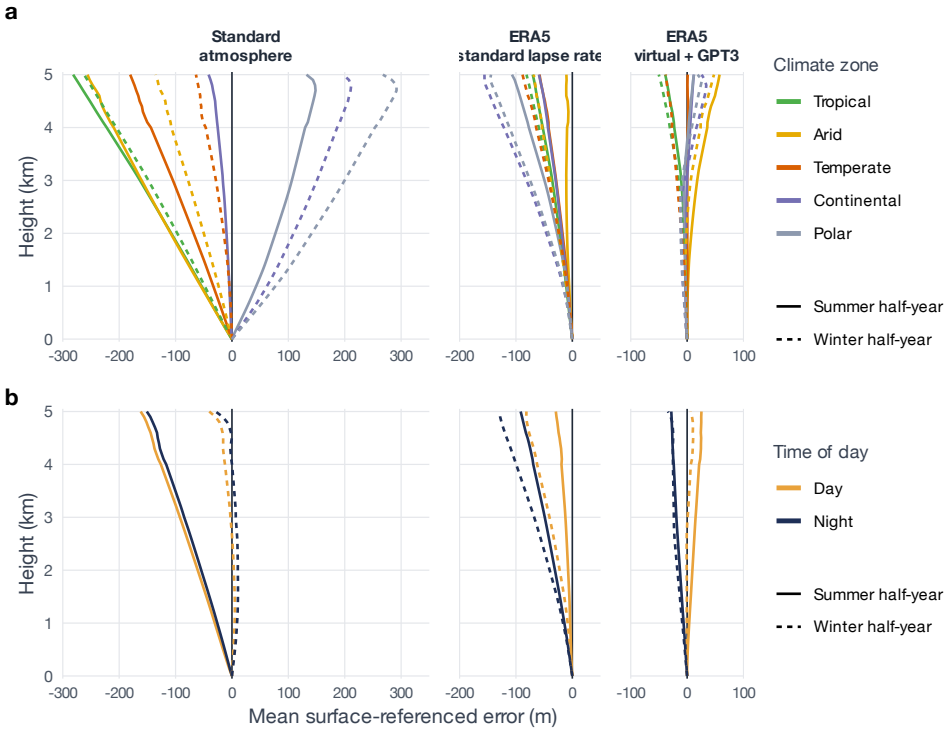


Figure 5: Mean signed flight-height error against height above the sounding surface for each conversion (columns), (a) by main Köppen–Geiger climate zone, coloured as in Figure 2, and (b) by day (07:00–19:00 local solar time) and night. Solid lines: summer half-year; dashed lines: winter half-year. All panels share the same horizontal scale; negative values indicate underestimated height.

rate removed most of this spread but still underestimated height in every climate zone and half-year, by 10 to 59 m, most in continental and polar winters. For this conversion, errors closely followed those predicted from the difference between assumed and observed mean column temperature ($r = 0.98$; Additional file 1: S2). With virtual temperature and GPT3, mean errors stayed within 13 m of zero in all climate zones and half-years. A day–night difference remained: in the summer half-year, the mean error was +12 m by day and -15 m at night, compared with -9 and -38 m with the standard lapse rate.

4 Discussion

Our results support a practical improvement to pressure-derived bird altitudes using openly available atmospheric information. Combining local ERA5 surface conditions with virtual temperature and a GPT3 lapse rate reduced median absolute ground error from approximately 38 to 4 m. Across the flight heights of tracked birds, median absolute atmospheric flight-height error fell from about 4% of height with the standard atmosphere to 1% with the full correction, and its 95th percentile from about 9% to 3%.

These reductions can help refine ecological patterns, although their ecological value depends on the vertical scale of the question. At a fixed ground elevation, weather-driven pressure changes can appear as gradual, continuous elevation changes when converted with a standard atmosphere (Figure 1). Correcting for weather helps separate this apparent movement from the timing and magnitude of actual elevational movements. In flight, a difference of a few tens of metres may leave a broad description of kilometre-high migration unchanged, yet matter when estimating exposure to turbine rotors [16, 33], responses to urban environments and artificial light [18, 45], or flight relative to topography [46]. For example, a radar experiment found a mean flight-height reduction of about 23 m under white-light treatment [18], illustrating that ecologically relevant responses can occur at this scale. Atmospheric errors also vary with climate, season and time of day (Figure 5), so comparisons of altitude among these conditions can inherit conversion differences.

Applying the correction requires no data beyond the pressure record, its timestamps and an approximate position. ERA5 provides hourly surface conditions worldwide [29, 37], GPT3 provides a global climatology of geographic and seasonal lapse-rate variation [34], and the GeoPressureR function `pressure_to_altitude()` combines them [47]. Because displacing the atmospheric query by tens of kilometres hardly changed the errors, approximate geolocator positions are sufficient.

The choice of atmospheric inputs nevertheless matters. ERA5-Land's finer grid produced larger persistent offsets at most GHCNh stations (Additional file 1: S3). This points to differences in the surface reference and supports using ERA5 single-level conditions for the evaluated conversion. For the atmospheric column, the virtual-temperature/GPT3 correction reduces recurring bias without requiring a temperature profile for every measurement. Its limited effect on within-station scatter is consistent with a climatology that cannot reproduce each day's conditions, including inversions. Lapse rates derived from ERA5 pressure levels, as used by Schaub et al. [31], or full atmospheric profiles, as evaluated for aviation [48], would capture such day-to-day departures and are expected to reduce errors aloft further. For year-round tracking records, however, retrieving pressure-level data for every measurement adds considerable data volume and processing time, which the small residual bias of the climatological correction may not justify.

Our results place earlier, largely assumed uncertainty statements on an empirical, height-resolved basis. For the standard atmosphere, they confirm the scale of previous estimates: the ground 95th percentile of 140 m agrees with pressure-system effects rarely exceeding 200 m [21], and a flight-height 95th percentile of about 9% of height agrees with the ±10% discussed by Sjöberg et al. [49]. They also show that these errors are not random scatter but depend systematically on climate, season and time of day (Figure 5); the underestimation that Lindström et al. [11] estimated for particular regional and seasonal conditions is one expression of this pattern. For weather-corrected altitudes, our results refine the ±100 m uncertainty commonly assumed [12, 20]. It is a generous bound at the median flight height, where the 95th percentile of atmospheric error was 19 m at 600 m, and still holds at the 90th percentile (69 m at 2,400 m), but is exceeded above about 3.2 km.

The remaining uncertainty depends on the quantity being interpreted. Persistent ground offsets can partly cancel in comparisons at one site, making relative elevation

changes less sensitive to those offsets than absolute elevations, although temporal variability remains. In rougher terrain, larger ground biases may reflect both atmospheric discrepancies and reference-height uncertainty, whose contributions cannot be separated here. A constant reference-height offset cannot explain the greater within-station SD. Sensor offset and drift add uncertainty beyond the atmospheric conversion; known-height events, such as floating periods in seabirds, can help calibrate records [16, 33]. Such calibration complements weather correction but does not determine the temperature profile aloft. Local ground clearance additionally requires terrain or sea-surface elevation in a consistent vertical reference [23, 50]. For example, the corrected conversion's 95th percentile of absolute error, 13 m at 100–250 m, is relative to the sounding surface and excludes surface-reference and sensor errors; it represents only part of the uncertainty relevant to a bird's position relative to a rotor or terrain feature.

The reference data limit how far these findings transfer to tracking records. Results depend on station selection: automatic pressure checks and terrain screening are independent of ERA5 residuals, but manual review included inspection of those residuals. Curation therefore does not provide an independent guarantee of reference quality and may remove genuine model discrepancies; the retained sample includes unresolved stable offsets (Additional file 1: S1.1). Reported radiosonde heights can likewise be derived from the measured atmospheric profile, so our flight comparison primarily tests the assumed atmospheric column rather than independently measured geometric height. ERA5 assimilates conventional surface and upper-air observations [29]; assimilation of the evaluated records was not determined. Balloon drift, uneven geographic coverage and the absence of offshore references further limit transfer. Within this scope, the results support using readily accessible weather and climatological information to reduce avoidable conversion errors, while selecting uncertainty summaries appropriate to the height, vertical reference and ecological question.

5 Conclusions

A fixed standard atmosphere introduces substantial and condition-dependent errors into pressure-derived bird altitudes. Referencing the conversion to local ERA5 surface conditions removes most of this error on the ground and in flight, and adding virtual temperature and a GPT3 lapse rate largely removes the remaining systematic flight-height bias. The correction needs only the pressure record, its timing and an approximate position, and is available in GeoPressureR, making it practical for long geolocator records. Day-to-day temperature profiles, terrain, sensor errors and position uncertainty still limit individual altitude estimates, particularly near ecological or turbine-height thresholds.

6 List of abbreviations

ARCO: analysis-ready, cloud-optimized; ECMWF: European Centre for Medium-Range Weather Forecasts; ERA5: fifth-generation ECMWF reanalysis; GHCNh: Global Historical Climatology Network hourly; GPS: Global Positioning System; GPT3: Global Pressure and Temperature 3; IGRA2: Integrated Global Radiosonde Archive, version 2; NOAA: National Oceanic and Atmospheric Administration; SD: standard deviation; UTC: Coordinated Universal Time.

7 Declarations

Ethics approval and consent to participate. This study reanalysed publicly available bird-tracking data and involved no new animal capture, handling or experimental procedures. The source dataset and associated studies provide the original collection information [44].

Consent for publication. Not applicable.

Availability of data and materials. The study code, derived tables and figures are available in the project repository [51]. The code and data archive has the reserved Zenodo identifier https://doi.org/10.5281/zenodo.23085093 [52]. The bird-height distribution uses the open-access Master GeoLocator Data Package, version 2026.0 [44], available at https://doi.org/10.5281/zenodo.18187093. The conversions are implemented in GeoPressureR [47]; the analysis uses version 3.6.6.9000 (f1431bc3). The study code is written in R and distributed under the MIT licence, without restrictions on commercial use; required packages and execution instructions are described in the project README [51]. GHCNh, IGRA2 and ERA5 data are openly available from their providers (Additional file 1: S1).

Competing interests. The authors declare that they have no competing interests.

Funding. This research received no dedicated funding.

Authors' contributions. RN designed the study, performed the analysis and wrote the manuscript. MG contributed to the implementation of the conversions in GeoPressureR and GeoPressureAPI and revised the manuscript.

Acknowledgements. Not applicable.

8 Additional files

Additional file 1. File name: `Additional_file_1.pdf`. File format: PDF (.pdf). Title: Supplementary methods and results. Description: Reference-station selection, environmental variation in conversion errors, the ERA5-Land comparison, the bird flight-height distribution, and sensitivity to horizontal-position uncertainty.

9 References

1. Nilsson C, Shamoun-Baranes J, Satterfield DA, Sjöberg S, Cohen EB. Animal niches in the airspace. Trends in Ecology & Evolution. 2025;40:1101–12. https://doi.org/10.1016/j.tree.2025.08.006.

2. Kemp MU, Shamoun-Baranes J, Dokter AM, Loon E van, Bouten W. The influence of weather on the flight altitude of nocturnal migrants in mid-latitudes. Ibis. 2013;155:734–49. https://doi.org/10.1111/ibi.12064.

3. Senner NR, Stager M, Verhoeven MA, Cheviron ZA, Piersma T, Bouten W. High-altitude shorebird migration in the absence of topographical barriers: Avoiding high air temperatures and searching for profitable winds. Proceedings of the Royal Society B: Biological Sciences. 2018;285:20180569. https://doi.org/10.1098/rspb.2018.0569.

4. Shamoun-Baranes J, Liechti F, Vansteelant WMG. Atmospheric conditions create freeways, detours and tailbacks for migrating birds. Journal of Comparative Physiology A. 2017;203:509–29. https://doi.org/10.1007/s00359-017-1181-9.

5. Dreelin RA, Shipley JR, Winkler DW. Flight behavior of individual aerial insectivores revealed by novel altitudinal dataloggers. Frontiers in Ecology and Evolution. 2018;6. https://doi.org/10.3389/fevo.2018.00182.

6. Williams HJ, Shepard ELC, Holton MD, Alarcón PAE, Wilson RP, Lambertucci SA. Physical limits of flight performance in the heaviest soaring bird. Proceedings of the National Academy of Sciences. 2020;117:17884–90. https://doi.org/10.1073/pnas.1907360117.

7. Dhanjal-Adams KL, Bauer S, Emmenegger T, Hahn S, Lisovski S, Liechti F. Spatiotemporal group dynamics in a long-distance migratory bird. Current Biology. 2018;28:2824–2830.e3. https://doi.org/10.1016/j.cub.2018.06.054.

8. Meier CM, Karaardıç H, Aymí R, Peev SG, Bächler E, Weber R, et al. What makes Alpine swift ascend at twilight? Novel geolocators reveal year-round flight behaviour. Behavioral Ecology and Sociobiology. 2018;72:45. https://doi.org/10.1007/s00265-017-2438-6.

9. Norevik G, Åkesson S, Andersson A, Bäckman J, Hedenström A. Flight altitude dynamics of migrating European nightjars across regions and seasons. Journal of Experimental Biology. 2021;224:jeb242836. https://doi.org/10.1242/jeb.242836.

10. Sjöberg S, Malmiga G, Nord A, et al. Extreme altitudes during diurnal flights in a nocturnal songbird migrant. Science. 2021;372:646–8. https://doi.org/10.1126/science.abe7291.

11. Lindström Å, Alerstam T, Andersson A, Bäckman J, Bahlenberg P, Bom R, et al. Extreme altitude changes between night and day during marathon flights of great snipes. Current Biology. 2021;31:3433–3439.e3. https://doi.org/10.1016/j.cub.2021.05.047.

12. Dufour P, Nussbaumer R, Briedis M, et al. Ecological barrier crossing strategies in small migratory birds depend on wing morphology and plumage color. iScience. 2026;29:114466. https://doi.org/10.1016/j.isci.2025.114466.

13. Barras AG, Liechti F, Arlettaz R. Seasonal and daily movement patterns of an alpine passerine suggest high flexibility in relation to environmental conditions. Journal of Avian Biology. 2021;52. https://doi.org/10.1111/jav.02860.

14. Rime Y, Nussbaumer R, Briedis M, et al. Multi-sensor geolocators unveil global and local movements in an Alpine-breeding long-distance migrant. Movement Ecology. 2023;11:19. https://doi.org/10.1186/s40462-023-00381-6.

15. Jensen FP, Werema C. Seasonal altitudinal migration of white-chested alethe chamaetylas fuelleborni in tanzania documented with a barometric pressure logger. Ostrich. 2025;96:69–73. https://doi.org/10.2989/00306525.2025.2481468.

16. Cleasby IR, Wakefield ED, Bearhop S, Bodey TW, Votier SC, Hamer KC. Three-dimensional tracking of a wide-ranging marine predator: Flight heights and vulnerability to offshore wind farms. Journal of Applied Ecology. 2015;52:1474–82. https://doi.org/10.1111/1365-2664.12529.

17. Doniol-Valcroze P, Beggs I, Bell F, et al. Multi-sensor logger tracking reveals the threats of offshore windfarms to passerine-sized birds. Conservation Letters. 2026;19:e70077. https://doi.org/10.1111/con4.70077.

18. Jimenez MF, Adams CA, Burt CS, Gerber BD, White J, Zinn R, et al. White light reduces flight height of birds in low-density migration corridor. The Journal of Wildlife Management. 2026;e70279. https://doi.org/10.1002/jwmg.70279.

19. Bäckman J, Andersson A, Alerstam T, Pedersen L, Sjöberg S, Thorup K, et al. Activity and migratory flights of individual free-flying songbirds throughout the

annual cycle: Method and first case study. Journal of Avian Biology. 2017;48:309–19. https://doi.org/10.1111/jav.01068.

20. Sjöberg S, Pedersen L, Malmiga G, et al. Barometer logging reveals new dimensions of individual songbird migration. Journal of Avian Biology. 2018;49:e01821. https://doi.org/10.1111/jav.01821.

21. Liechti F, Bauer S, Dhanjal-Adams KL, Emmenegger T, Zehtindjiev P, Hahn S. Miniaturized multi-sensor loggers provide new insight into year-round flight behaviour of small trans-Sahara avian migrants. Movement Ecology. 2018;6:19. https://doi.org/10.1186/s40462-018-0137-1.

22. Clark B, Handby T, Leat E, Weber S. First three-dimensional tracks for the ascension frigatebird fregata aquila highlight the importance of altitude for behavioural studies. Seabird Journal. 2019;32:1–17. https://doi.org/10.61350/sbj.32.1.

23. Péron G, Calabrese JM, Duriez O, et al. The challenges of estimating the distribution of flight heights from telemetry or altimetry data. Animal Biotelemetry. 2020;8:5. https://doi.org/10.1186/s40317-020-00194-z.

24. ICAO. Manual of the ICAO standard atmosphere, doc 7488/3. 3rd edition. Montreal: International Civil Aviation Organization; 1993.

25. Dhanjal-Adams KL, Willener AST, Liechti F. Pamlr: A toolbox for analysing animal behaviour using pressure, acceleration, temperature, magnetic or light data in R. Journal of Animal Ecology. 2022;91:1345–60. https://doi.org/10.1111/1365-2656.13695.

449 26. Lathouwers M, Artois T, Dendoncker N, Beenaerts N, Conway G, Henderson I, et
450 al. Rush or relax: Migration tactics of a nocturnal insectivore in response to ecological
451 barriers. Scientific Reports. 2022;12:4964. https://doi.org/10.1038/s41598-022-09106-y.

452 27. Zhao T, Heim W, Nussbaumer R, van Toor M, Zhang G, Andersson A, et al.
453 Seasonal migration patterns of Siberian Rubythroat (*Calliope calliope*) facing the
454 Qinghai–Tibet Plateau. Movement Ecology. 2024;12:54. https://doi.org/10.1186/s404
455 62-024-00495-5.

456 28. Brodersen JE, Lindström Å, Alerstam T, Andersson A, Bäckman J, Hedenström
457 A, et al. Low cost of shifting flight altitude in migrating great snipes. Journal of
458 Experimental Biology. 2026;229:jeb252022. https://doi.org/10.1242/jeb.252022.

459 29. Hersbach H, Bell B, Berrisford P, et al. The ERA5 global reanalysis. Quarterly
460 Journal of the Royal Meteorological Society. 2020;146:1999–2049. https://doi.org/10.1
461 002/qj.3803.

462 30. Nussbaumer R, Gravey M, Briedis M, Liechti F. Global positioning with animal-
463 borne pressure sensors. Methods in Ecology and Evolution. 2023;14:1104–17. https:
464 //doi.org/10.1111/2041-210X.14043.

465 31. Schaub T, Millon A, De Zutter C, et al. How to improve the accuracy of height
466 data from bird tracking devices? An assessment of high-frequency GPS tracking
467 and barometric altimetry in field conditions. Animal Biotelemetry. 2023;11:31. https:
468 //doi.org/10.1186/s40317-023-00342-1.

469 32. Lato KA, Stepanuk JEF, Heywood EI, Conners MG, Thorne LH. Assessing the accu-
470 racy of altitude estimates in avian biologging devices. PLOS ONE. 2022;17:e0276098.
471 https://doi.org/10.1371/journal.pone.0276098.

472 33. Johnston DT, Thaxter CB, Boersch-Supan PH, et al. Flight heights obtained
473 from GPS versus altimeters influence estimates of collision risk with offshore wind
474 turbines in lesser black-backed gulls *larus fuscus*. Movement Ecology. 2023;11:66.
475 https://doi.org/10.1186/s40462-023-00431-z.

476 34. Landskron D, Böhm J. VMF3/GPT3: Refined discrete and empirical troposphere
477 mapping functions. Journal of Geodesy. 2018;92:349–60. https://doi.org/10.1007/s001
478 90-017-1066-2.

479 35. Wallace JM, Hobbs PV. Atmospheric thermodynamics. In: Atmospheric science:
480 An introductory survey. 2nd edition. Academic Press; 2006. p. 63–111. https://doi.or
481 g/10.1016/B978-0-12-732951-2.50008-9.

482 36. Bolton D. The computation of equivalent potential temperature. Monthly Weather
483 Review. 1980;108:1046–53. https://doi.org/10.1175/1520-0493(1980)108%3C1046:
484 TCOEPT%3E2.0.CO;2.

485 37. Copernicus Climate Change Service. ERA5 hourly data on single levels from 1940
486 to present. 2018. https://doi.org/10.24381/cds.adbb2d47.

487 38. NOAA National Centers for Environmental Information. Global historical
488 climatology network hourly (GHCNh). 2026.

489 39. Durre I, Vose RS, Wuertz DB. Overview of the Integrated Global Radiosonde
490 Archive. Journal of Climate. 2006;19:53–68. https://doi.org/10.1175/JCLI3594.1.

491 40. Durre I, Yin X, Vose RS, et al. Enhancing the data coverage in the Integrated Global
492 Radiosonde Archive. Journal of Atmospheric and Oceanic Technology. 2018;35:1753–70.
493 https://doi.org/10.1175/JTECH-D-17-0223.1.

494 41. NOAA National Centers for Environmental Information. Integrated global
495 radiosonde archive: Product and sounding-format documentation.

496 42. Bryant C, Wheeler NR, Rubel F, French RH. Kgc: Koeppen-geiger climatic zones.
497 2017. https://doi.org/10.32614/CRAN.package.kgc.

498 43. Muñoz-Sabater J, Dutra E, Agustí-Panareda A, et al. ERA5-Land: A state-of-
499 the-art global reanalysis dataset for land applications. Earth System Science Data.
500 2021;13:4349–83. https://doi.org/10.5194/essd-13-4349-2021.

501 44. Nussbaumer R, Adamik P, Alerstam T, Andersson A, Arlettaz R, Bächler E, et al.
502 Master GeoLocator data package. 2026. https://doi.org/10.5281/zenodo.18187093.

503 45. Cabrera-Cruz SA, Smolinsky JA, McCarthy KP, Buler JJ. Urban areas affect flight
504 altitudes of nocturnally migrating birds. Journal of Animal Ecology. 2019;88:1873–87.
505 https://doi.org/10.1111/1365-2656.13075.

506 46. Bishop CM, Spivey RJ, Hawkes LA, Batbayar N, Chua B, Frappell PB, et al. The
507 roller coaster flight strategy of bar-headed geese conserves energy during himalayan
508 migrations. Science. 2015;347:250–4. https://doi.org/10.1126/science.1258732.

509 47. Nussbaumer R, Gravey M. GeoPressureR: Global positioning by atmospheric
510 pressure. 2022. https://doi.org/10.5281/zenodo.7754457.

511 48. Simonetti M, García Crespillo O. Geodetic altitude from barometer and weather
512 data for GNSS integrity monitoring in aviation. NAVIGATION. 2024;71:navi.637.
513 https://doi.org/10.33012/navi.637.

514 49. Sjöberg S, Andersson A, Bäckman J, Hansson B, Malmiga G, Tarka M, et al. Solar
515 heating may explain extreme diel flight altitude changes in migrating birds. Current
516 Biology. 2023;33:4232–4237.e2. https://doi.org/10.1016/j.cub.2023.08.035.

517 50. Poessel SA, Duerr AE, Hall JC, Braham MA, Katzner TE. Improving estimation of
518 flight altitude in wildlife telemetry studies. Journal of Applied Ecology. 2018;55:2064–70.
519 https://doi.org/10.1111/1365-2664.13135.

520 51. Nussbaumer R, Gravey M. Altitude validation: Study code, derived tables and
521 figures. 2026.

522 52. Nussbaumer R, Gravey M. How accurate is altitude from pressure geolocators?
523 Code, data and report. 2026. https://doi.org/10.5281/zenodo.23085093.